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Multi-omics analysis for identifying cell-type-specific and bulk-level druggable targets in Alzheimer's disease

Shiwei Liu^{1,2}, Minyoung Cho^{1,2,3}, Yen-Ning Huang^{1,2}, Tamina Park^{1,2}, Soumilee Chaudhuri^{1,2}, Thea J. Rosewood^{1,2}, Paula J. Bice^{1,2}, Dongjun Chung⁴, David A. Bennett⁵, Nilüfer Ertekin-Taner^{6,7}, Andrew J. Saykin^{1,2} and Kwangsik Nho^{1,2,8*} 

Abstract

Background Analyzing disease-linked genetic variants via expression quantitative trait loci (eQTLs) helps identify potential disease-causing genes. Previous research prioritized genes by integrating Genome-Wide Association Study (GWAS) results with tissue-level eQTLs. Recent studies have explored brain cell type-specific eQTLs, but a systematic analysis across multiple Alzheimer's disease (AD) genome-wide association study (GWAS) datasets or comparisons between tissue-level and cell type-specific effects remain limited. Here, we integrated brain cell type-level and bulk-level eQTL datasets with AD GWAS datasets to identify potential causal genes.

Methods We used Summary Data-Based Mendelian Randomization (SMR) and Bayesian Colocalization (COLOC) to integrate AD GWAS summary statistics with eQTLs datasets. Combining data from five AD GWAS, two single-cell eQTL datasets, and one bulk eQTL dataset, we identified novel candidate causal genes and further confirmed known ones. We investigated gene regulation through enhancer activity using H3K27ac and ATAC-seq data, performed protein–protein interaction (PPI) and pathway enrichment, and conducted a drug/compound enrichment analysis with Drug Signatures Database (DSigDB) to support drug repurposing for AD.

Results We identified 28 candidate causal genes for AD, of which 12 were uniquely detected at the cell-type level, 9 were exclusive to the bulk level and 7 detected in both. Among the 19 cell-type level candidate causal genes, microglia contributed the highest number of candidate genes, followed by excitatory neurons, astrocytes, inhibitory neurons, oligodendrocytes, and oligodendrocyte precursor cells (OPCs). *PABPC1* emerged as a novel candidate causal gene in astrocytes. We generated PPI networks for the candidate causal genes and found that pathways such as membrane organization, cell migration, and ERK1/2 and PI3K/AKT signaling were enriched. The AD-risk variant associated with candidate causal gene *PABPC1* is located near or within enhancers only active in astrocytes. We classified the 28 genes into three drug tiers and identified druggable interactions, with imatinib mesylate emerging as a key candidate. A drug-target gene network was created to explore potential drug targets for AD.

Conclusions We systematically prioritized AD candidate causal genes based on cell type-level and bulk level molecular evidence. The integrative approach enhances our understanding of molecular mechanisms of AD-related genetic variants and facilitates interpretation of AD GWAS results.

*Correspondence:

Kwangsik Nho

knho@iu.edu

Full list of author information is available at the end of the article



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Keywords Causal genes, eQTL, Alzheimer's disease, GWAS, SNP, Genetic variant, Gene expression, Cell type, Astrocytes, Drug repurposing

Background

Alzheimer's Disease (AD) is a multifaceted neurodegenerative disorder characterized by progressive cognitive decline and memory loss [1]. AD is broadly categorized into early-onset and late-onset forms, with late-onset AD (LOAD) being the most common [2]. The genetic architecture of AD is complex, involving numerous deleterious variants distributed across various genes [2]. Among these, the *APOE* $\epsilon 4$ allele is recognized as the strongest genetic risk factor for late-onset AD [3]. Genome-Wide Association Studies (GWAS) have significantly advanced our understanding of the genetic basis of AD [4–9]. Early AD GWAS studies identified key loci like *CLU* and *CRI* [5]. The latest AD GWAS study has significantly expanded our understanding of the genetic basis of Alzheimer's disease, identifying 83 genetic variants across 75 loci, including 42 newly discovered variants in European ancestry populations [4].

However, while GWAS studies are instrumental in identifying genetic variants associated with AD, they fail to elucidate the molecular and cellular mechanisms by which the variants contribute to the disease. Only a small fraction of these variants resides within coding regions, while a significant number of non-coding risk variants remain unexplained. To better understand the underlying mechanisms through which these risk variants act, recent studies have employed Expression Quantitative Trait Loci (eQTL) analyses [10–15] for following up GWAS study results. The eQTL analyses can reveal how non-coding variants identified by GWAS influence the risk of AD through changes in gene expression [16]. Several public eQTL datasets derived from brain tissue have become available, including the Braineac dataset from the UK Brain Expression Consortium (UKBEC) [17], the Genotype-Tissue Expression (GTEx) consortium [18] and the MetaBrain dataset [10]. These datasets have enhanced the interpretation of GWAS findings by elucidating how risk variants regulate gene expression on the tissue level.

Furthermore, a few recent eQTL studies have demonstrated that these non-coding variants affect gene expression in a cell-type-specific manner, underscoring the complexity of their functional impact [11, 14]. Cell type-specific eQTLs enable researchers to determine the cell types that are most influenced by genetic variants and enable the identification of key cell types and regulatory networks involved in the disease progression, thereby offering enhanced understanding of the

underlying mechanisms of diseases. Moreover, previous research has shown that GWAS-identified risk variants in non-coding regions can influence phenotypic outcomes by perturbing transcriptional gene promoters and enhancers [19]. For instance, a study has shown that the AD-associated genes *BIN1* and *PICALM* are regulated by AD risk variants that overlap with microglia-specific enhancers, which interact with the active promoters of these genes [19]. Understanding whether these genetic risk variants overlap with specific regulatory elements provides deeper insights into the cell-type-specific mechanisms underlying gene expression regulation.

In this study, we systematically integrated AD GWAS summary statistics with cell type-level and bulk-level eQTL data to enhance our understanding of the genetic mechanisms underlying AD. We employed Summary Data-Based Mendelian Randomization (SMR) and Bayesian colocalization (COLOC) methods to identify and prioritize potential disease-causing genes. Our analysis included five recent AD GWAS datasets and two cell type-level eQTL datasets derived from single-cell sequencing of AD brain samples, as well as a tissue-level metabrain eQTL dataset from previous studies. We focused on prioritizing candidate causal genes for follow-up functional studies in the future. We examined their associated variants and the possible effects on enhancers in a cell type-specific manner. By comparing our results with existing studies, we identified novel cell type-specific candidate genes and used tools such as eQTLot to visualize their colocalization. Additionally, we used differential gene expression analysis data to investigate the associations between these novel candidate causal genes and AD. This comprehensive approach aims to improve our understanding of AD's genetic basis at the molecular and cellular level and identify potential therapeutic targets.

Methods

Datasets

We utilized summary statistics from 5 latest GWAS studies on AD involving European ancestry, downloaded from the NHGRI-EBI GWAS Catalog. As shown in Additional file 1: Table S1, Kunkle et al. [6] included 21,982 AD cases, and 41,944 controls from the U.S., Canada, France, Germany, Netherlands, Iceland, U.K., Greece, and other regions, totaling 63,926 samples. Jansen et al. [7] involved 24,087 AD cases, 47,793 proxy cases, and

383,378 controls, with a total of 455,258 samples from the U.S., Norway, Sweden, U.K., and other regions (Additional file 1: Table S1). Wightman et al. [9] analyzed 39,918 AD cases, 46,613 proxy cases, and 676,386 controls, with a total sample size of 762,917 from Finland, Iceland, Norway, Spain, Sweden, U.K., U.S., and other regions (Additional file 1: Table S1). Schwartzentruber et al. [8] included 21,982 AD cases, 53,000 proxy cases, and 419,944 controls, totaling 472,868 samples from Greece, Canada, U.S., U.K., France, and Germany (Additional file 1: Table S1). Bellenguez et al. [4] provided data on 39,106 clinically diagnosed AD cases, 46,828 proxy cases, and 401,577 controls, amounting to 487,511 samples from Portugal, Switzerland, Spain, Greece, Czech Republic, Netherlands, Sweden, U.S., Belgium, Norway, Finland, Denmark, Italy, U.K., Bulgaria, France, and Germany (Additional file 1: Table S1).

We utilized multiple cis-eQTL datasets predominantly from individuals of European ancestry, including both tissue and cell type levels datasets in brain cortex (Additional file 1: Table S2). The Metabrain eQTL dataset provides tissue-level cis-eQTL data derived from a meta-analysis of 14 bulk RNA-seq datasets focused on the brain cortex [10] (see Additional file 1: Table S2). Two cell type-specific cis-eQTL datasets were obtained from single-cell sequencing data. Bryois et al. [11] provided a cell type-specific eQTL dataset encompassing the temporal cortex, white matter, DLPFC, and prefrontal cortex (PFC) (see Additional file 1: Table S2). Moreover, we performed eQTL analysis and generated a cell type-specific eQTL dataset utilizing the snRNA data from the DLPFC region of the Religious Orders Study and Memory and Aging Project (ROSMAP) cohort, as reported in the study by Mathys et al. [20] (Additional file 1: Table S2).

eQTL analysis

To conduct eQTL analysis using the snRNA dataset from the Mathys et al. [20] ROSMAP cohort, we generated pseudobulk expression profiles. We focused on seven main cell types (Excitatory neurons, Inhibitory neurons, Oligodendrocytes, Oligodendrocyte Progenitor Cells (OPCs), Astrocytes, Immune cells, Vasculature cells). Pseudobulk UMI count matrices for each cell type were generated by summing UMI counts per gene across all cells within each individual using Seurat (Version 5.0.1). Low-expression genes were filtered out using the 'filterByExpr' function from edgeR (version 3.40.2) with default parameters. The remaining pseudobulk counts were normalized using the trimmed mean of M-values (TMM) method from edgeR, and log2 counts per million (CPM) were computed and then quantile normalized

with the 'voom' function from limma (version 3.54.2) as a previous study [14].

To identify cis-eQTLs within 1 Mb of the transcription start site of each gene, we used Matrix EQTL (version 2.3) for analysis. Bi-allelic SNPs were retained if they had a minor allele frequency > 0.05, a call rate > 95%, and Hardy–Weinberg equilibrium $p > 10^{-6}$ using PLINK2 as a previous study [14]. Gene expression was modeled using a linear regression with SNP allele counts and several covariates, and significance was determined by t-statistics. To account for population structure, the top 3 genotype principal components (PCs) were included as covariates as a previous study [14]. Additionally, the top 40 expression PCs, calculated within each cell type, were chosen because they captured, on average, 95% of the expression variance across cell types, controlling for non-genetic sources of variation. Covariates including age, sex, post-mortem interval, study cohort (ROS or MAP), and total number of genes detected were also included as a previous study [14].

Summary data-based Mendelian randomization

We performed SMR and Heterogeneity in Dependent Instruments (HEIDI) tests to investigate pleiotropic associations between gene expression and AD within cis-regions, using the SMR software tool (version 1.3.1). The SMR method, as detailed in the previous study [21], enables the testing of whether the effect size of a SNP on a phenotype is mediated through gene expression. This tool facilitates the prioritization of candidate causal genes underlying GWAS hits for further functional studies by leveraging summary-level data from both GWAS and eQTL datasets (as mentioned above). For our analysis, we used default parameters in the SMR software with a p-value threshold of 5.0×10^{-8} to select the top associated eQTLs for the SMR test, focusing exclusively on cis-regions. The HEIDI test, which assesses heterogeneity among dependent instruments, was conducted using a default eQTL p-value threshold of 1.57×10^{-3} to filter SNPs for each probe, corresponding to a chi-squared value (df=1) of 10. The association between gene expression and AD was determined as P-value of $\text{SMR} < 0.05 / \text{number of probes tested}$. For the HEIDI test, significance was determined as P-value of $\text{HEIDI} > 0.05$ as previous studies [21].

Bayesian colocalization analysis

We conducted colocalization analysis using the Coloc package (version 5.2.3) [22] to investigate whether AD phenotype and gene expression share common causal variants in a given region. The input data comprised SNP effect sizes and associated p-values from both the AD GWAS and eQTL datasets (as mentioned above),

formatted according to the package's requirements. Using the `coloc.abf` function in the package, we tested the hypothesis of a shared causal variant, assuming at most one causal variant per trait. Colocalization analysis calculates posterior probabilities (PPs) of the five hypotheses: (1) PPH0; no association with either gene expression or phenotype; (2) PPH1; association with gene expression, not with the phenotype; (3) PPH2; association with the phenotype, not with gene expression; (4) PPH3; association with gene expression and phenotype by independent SNVs; and (5) PPH4; association with gene expression and phenotype by shared causal SNVs. As a large PP for H4 strongly supports shared causal variants affecting both gene expression and phenotype, we considered $PPH4 > 0.75$ and $PPH4/PPH3 > 3$ as strong evidence for colocalization as previous studies [23].

Visualization of identified candidate causal genes across eQTL and GWAS datasets

Figures S1–S5 in Additional file 2 present heatmaps displaying SMR beta values and significance levels for candidate causal genes identified through SMR and colocalization analyses for five AD GWAS datasets respectively. For genes detected in multiple GWAS datasets, the direction of effect (MR beta sign) was found to be consistently aligned across the GWAS datasets. To summarize and integrate findings across all five GWAS datasets, we generated a comprehensive heatmap showing the sign of SMR beta values for each candidate gene, including genes found to be significant in at least one GWAS analysis. These heatmaps were generated using the ComplexHeatmap R package [24], with columns grouped by eQTL datasets (two cell-type-level and one bulk-level) and rows representing candidate genes.

Network analysis of candidate causal genes

We utilized the identified candidate causal genes as input to construct a protein–protein interaction (PPI) network. This network was generated using STRING (version 12.0) with a confidence score threshold of 0.4 as the minimum required interaction score and default settings for all other parameters. The resulting network was then visualized using Cytoscape (version 3.10.2). In the network, nodes represent proteins, while edges illustrate the interactions between these molecules. The shape of each node in the PPI network was used to represent the detection context of candidate genes: ellipses indicate genes detected only in cell type-level datasets, diamonds represent genes found only in bulk-level datasets, and rectangles denote genes shared between both cell type-level and bulk-level analyses. The thickness of the edges corresponds to the combined interaction score from STRING.

Pathway enrichment of all candidate causal genes

To identify biologically relevant pathways and processes associated with all candidate causal mRNA genes, we performed functional enrichment analysis using the `enrichR` R package [25]. We conducted enrichment analysis with multiple up-to-date databases, including KEGG 2021 Human, Reactome Pathways 2024, WikiPathways 2024 Human, GO Biological Process 2025, GO Cellular Component 2025, GO Molecular Function 2025. Enrichment results were filtered to retain only those terms with an adjusted P-value ≤ 0.05 and a minimum gene count of 3. The GeneRatio was computed as the number of overlapping genes (Gene Count) divided by the total number of genes in each term (Total In Term). The significantly enriched terms were ranked by GeneRatio values and visualized using the `ggplot2` package in R.

eQTLplot analysis for visualizing colocalization

We utilized the eQTLplot (version 0.0.0.9000) R package to visualize the colocalization between AD GWAS data and eQTL data [26]. This tool enables comprehensive visualization of gene-trait interactions by generating a series of customizable plots. Using eQTLplot, we produced visualizations that highlight the overlap between AD GWAS and eQTL signals, the correlation between their p-values, and the enrichment of eQTLs among trait-significant variants. Additionally, the tool provided insights into the linkage disequilibrium (LD) landscape of the locus and the relationship between the directions of effect for eQTL signals and colocalizing GWAS peaks, which help us to better understand the genetic relationships between gene expression and AD.

Cell-type-specific enhancer activity analysis

GWAS risk variants located in noncoding regions can influence phenotypic outcomes by affecting transcriptional gene promoters and enhancers [19]. Clusters of enhancers, known as super-enhancers, play a vital role in regulating cell-identity genes and are key to establishing cell-type-specific gene expression patterns [19]. In this study, we evaluated the impact of disease variants on cis-gene expression in specific cell types by evaluating whether disease variants are located within or next to regulatory elements, including enhancers and promoters. A previous study highlights that while active promoters are typically conserved across different brain cell types, active enhancers exhibit marked cell-type specificity [19]. Thus, we focused on variant-enhancer analysis. We used a publicly available dataset, including ATAC-seq, which identifies open chromatin regions, and ChIP-seq, which marks active enhancers (H3K27ac) and promoters (H3K4me3) for each brain cell type, accessed through

the UCSC genome browser session (hg19). This dataset was generated from nuclei isolated from brain tissue resected during epilepsy treatment in 10 individuals [19]. This approach allowed us to identify which enhancers are active in specific cell types, thereby elucidating the cell-type-specific effects of disease variants on gene expression.

Druggability analysis

To identify druggable genes, we classified all identified candidate causal genes into three tiers based on druggability confidence according to a previous study [27]. Tier 1 included genes whose protein products are targets of approved small molecule, and biotherapeutic drugs were identified using manually curated efficacy target information from release 17 of the ChEMBL database. Tier 2 comprised proteins with high sequence similarity to drug targets or proteins associated with drug-like compounds, identified through a BLASTP search of Ensembl peptide sequences against approved drug efficacy targets. Tier 3 encompassed proteins with more distant relationships to drug targets, identified by BLASTP with $\geq 25\%$ identity over $\geq 75\%$ of the sequence and $E\text{-value} \leq 0.001$. Additionally, to prioritize alternative targets for non-druggable candidate causal genes, we utilized data from EpiGraphDB to identify directly AD related interacting genes that are indicated to be druggable with Tier1 druggability [28] based on protein–protein interaction (PPI) networks (IntAct [29], STRING [30]) and with literature or xQTL evidence for AD relevance [28].

Potential drug/compound prediction

To identify potential pharmacological drug/compound that could modulate the expression of candidate causal genes for AD, we utilized the Drug Signatures Database (DSigDB) [31]. This resource includes 22,527 gene sets and 17,389 unique compounds linked to 19,531 genes. We accessed and downloaded the annotated drug/compound gene sets from DSigDB's official website [32]. Using the enricher function from the R package clusterProfiler (version 4.10.1), we performed enrichment analysis to explore connections between our target genes (either druggable causal genes or tier 1 interacting genes) and potential drugs, aiming for AD drug repurposing. We set an Benjamini–Hochberg adjusted p-value threshold of < 0.01 to identify drugs significantly associated with these target genes. The top 10 enriched drugs/compounds were visualized using a dot plot, and an interaction network was generated with Cytoscape (version 3.10.2) to illustrate the relationships between the target genes and the enriched drugs/compounds.

Results

Workflow

To identify and prioritize genes associated with AD, we integrated summary-level data from GWAS with eQTL data. As shown in Fig. 1, we incorporated data from five recent AD GWAS datasets and two cell type-level eQTL datasets from single-cell sequencing of AD brain samples, along with a tissue-level Metabrain eQTL dataset from previous research, as described in “Methods”. As outlined in Fig. 1, we first employed SMR to evaluate how SNPs associated with AD risk influence gene expression. Subsequently, we used Coloc to validate the colocalization of genetic variants within specific genomic regions. We identified 28 candidate causal genes that met our rigorous criteria (Fig. 2). We explored how associated variants might regulate gene expression in a cell type-specific manner, utilizing previous data on cell type-specific enhancers or promoters in brain tissue. Additionally, we compared our findings with prior studies to highlight novel candidate genes with less previous support as shown in Fig. 1. For these novel genes, we visualized colocalization results and derived differential gene expression data from earlier studies to confirm their association with AD. Finally, we assessed the druggability of the prioritized candidate causal genes to explore potential therapeutic targets.

Summary results of detected candidate causal genes

We integrated data from five recent AD GWAS datasets and two cell type-level eQTL datasets obtained from single-cell sequencing of AD brain samples, along with a metabrain tissue-level eQTL dataset from prior research (Fig. 2). The Bryois single cell RNA data was from a range of regions such as the temporal cortex, white matter, and PFC, while the Mathys et al. single cell RNA sequencing data was from DLPFC region (Fig. 2). Utilizing SMR and HEIDI as well as Coloc analyses, we identified 28 candidate causal genes across these datasets that met the filtering criteria: SMR $FDR < 0.05$, HEIDI p-value > 0.05 , Coloc $PPH4 > 0.75$, and Coloc $PPH4/PPH3 > 3$, as shown in Fig. 2 and Additional file 1: Tables S3–S5. Out of the 28 candidate causal genes, two (AL355353.1 and AL137789.1) are lncRNA genes, while the remaining 26 genes are mRNA genes. As shown in Additional file 2: Figures S1–S5, the Bellenguez AD GWAS summary statistics revealed the highest number of candidate causal genes compared to the other AD GWAS datasets. Among the 28 candidate causal genes, 12 were uniquely detected at the cell-type level, 7 were shared between cell-type and bulk analyses, and 9 were exclusive to the bulk level (Additional file 2: Figure S6. Furthermore, we noted concordant MR beta signs between cell type-level eQTL and bulk-level eQTL datasets. *EGFR*, *SNX31*, *PABPC1*,

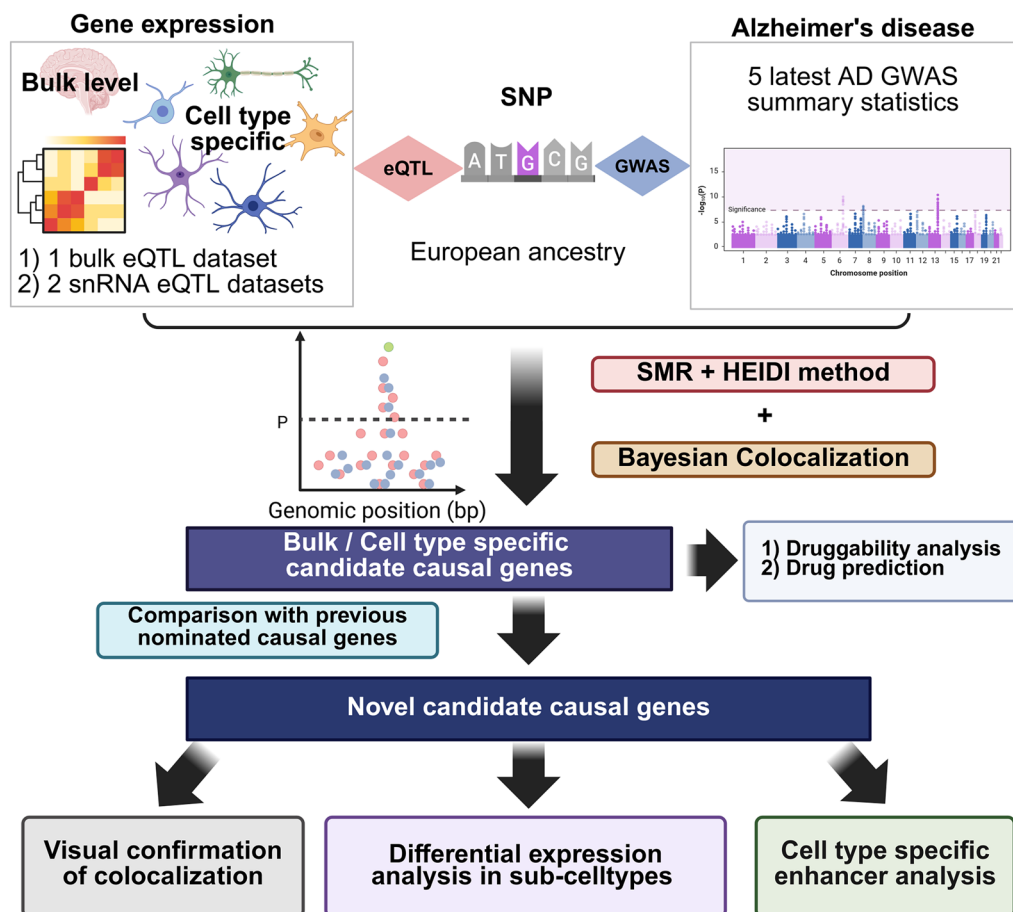


Fig. 1 Study workflow. Created by BioRender

ACE, *ARL17B*, *PRSS36* and *LRRC37A* were identified in both the bulk-level and cell type-level datasets with aligned direction of effect (Fig. 2, Additional file 2: Figure S6). Additionally, *TSPAN14*, *GRN*, *CD2AP*, *APH1B*, *SLC39A13*, *FCER1G*, *CR1*, *NDUFAF6*, *TP53INP1* were identified exclusively as candidate causal genes in the bulk metabrain eQTL dataset (Fig. 2, Additional file 2: Figure S6). *RIN3*, *PICALM*, *JAZF1*, *RABEP1*, *KANSL1*, *AL355353.1*, *SCIMP*, *USP6NL*, *CASS4*, *FERMT2*, *BIN1*, *ALI37789.1* were identified exclusively as candidate causal genes in the cell type eQTL datasets (Fig. 2, Additional file 2: Figure S6).

With the combined results from all GWAS datasets, of the 19 cell type level candidate causal genes, 16 were found to be causal in only one cell type (Fig. 2 and Table 1). While 3 genes including *ACE*, *ARL17B* and *SCIMP* were shared across multiple cell types with concordant MR beta signs, as shown in Fig. 2 and Table 1. The highest number of candidate causal genes was detected in microglia, followed by excitatory neurons, astrocytes, inhibitory neurons, oligodendrocytes, and

OPCs (Fig. 2 and Table 1). We identified 5 cell type-specific candidate causal genes (*EGFR*, *SNX31*, *PICALM*, *JAZF1* and *RABEP1*), which were detected in both snRNA eQTL datasets (Table 1).

To evaluate the novelty and robustness of candidate causal genes identified in this study, we cross-referenced them with prior literature [4, 6–8, 10, 11, 14, 15, 33, 34]. Genes were grouped based on whether they have support from previous bulk-level or cell type-specific studies (Agora nomination or significant detection by SMR or Coloc analysis) or represent novel findings. Firstly, for the 9 candidate causal genes exclusively detected in the bulk-level dataset, all of them were previously reported in the literature [4, 6, 10, 15, 33] (Table 2). Interestingly, *FCER1G*, *GRN*, *CD2AP*, *APH1B*, and *CR1* have also been identified as cell type-level causal genes in previous studies by colocalization or SMR analyses [11, 14] (Table 2). Secondly, for the 12 genes identified exclusively in cell type-level datasets, genes including *RIN3*, *BIN1*, *CASS4*, *RABEP1*, *AL355353.1*, *SCIMP*, *USP6NL*, *PICALM* have

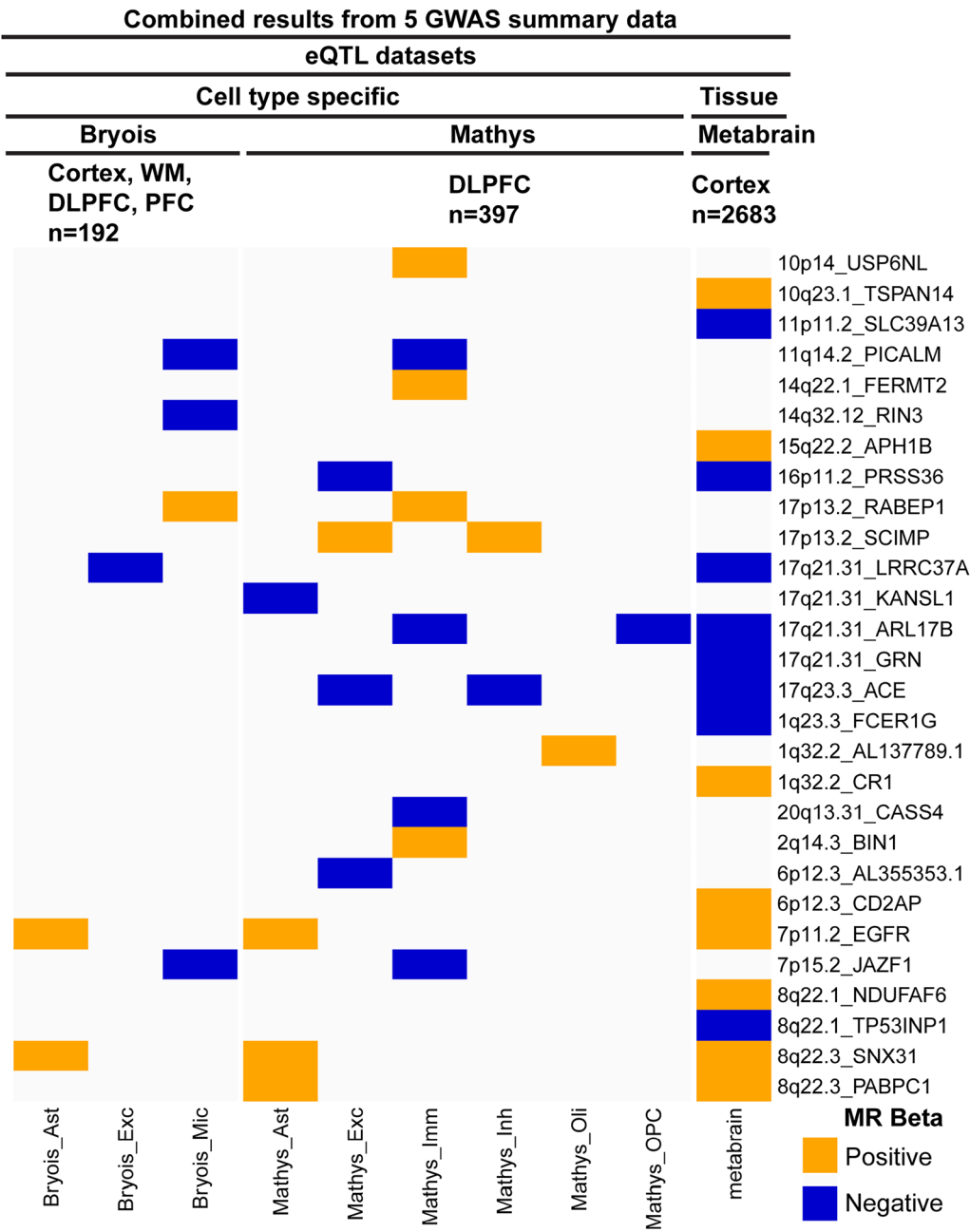


Fig. 2 SMR beta value signs for candidate causal genes from SMR and colocalization analysis. Note: all five GWAS datasets results are combined. The candidate causal genes are filtered by SMR FDR < 0.05, HEIDI > 0.05, Coloc PPH4 > 0.75, Coloc PPH4/PPH3 > 3

been previously supported by both bulk and cell type-level evidence [4, 6–8, 10, 11, 14, 15, 33, 34] (Table 2). *KANSL1*, *AL137789.1* and *FERMT2* have prior support at the bulk-level [4, 6, 10, 15, 33] and *JAZF1* has prior support at the cell type level [11, 14] (Table 2). Additionally, the previous known candidate causal gene *AL137789.1* and *FERMT2* at the bulk level are now confirmed with new cell type-level evidence in this

study (Table 2). Lastly, for the 7 genes identified in both cell type-level and bulk-level datasets, *ACE* and *PRSS36* have been previously supported by both bulk and cell type-level evidence [4, 6–8, 10, 11, 14, 15, 33, 34] (Table 2). *EGFR*, *SNX31*, *ARL17B* and *LRRC37A* have prior support at the bulk-level [4, 6, 10, 15, 33] and are now confirmed with new cell type-level evidence in this study (Table 2). Lastly, *PABPC1* emerged as a

Table 1 Comparison of cell type-level candidate causal genes between the two snRNA eQTL datasets

Celltypes	Cell type-level causal genes (combined results with 5 GWAS summary data)		
	Uniquely identified in Bryois snRNA dataset	Uniquely identified in the Mathys snRNA dataset	Identified in both snRNA datasets
Astrocytes	–	<i>PABPC1, KANSL1</i>	<i>EGFR, SNX31</i>
Excitatory Neurons	<i>LRRC37A</i>	<i>PRSS36, AL355353.1, ACE, SCIMP</i>	–
Immune Cells or Microglia	<i>RIN3</i>	<i>USP6NL, FERMT2, ARL17B, CASS4, BIN1</i>	<i>PICALM, JAZF1, RABEP1</i>
Inhibitory Neurons	–	<i>SCIMP, ACE</i>	–
Oligodendrocytes	–	<i>AL137789.1</i>	–
OPCs	–	<i>ARL17B</i>	–

The underlined genes are identified as candidate causal genes in multiple cell types

Table 2 Comparison of candidate causal genes identified in this study with prior literature

Detection categories	Gene groups based on literature support	Gene names
Detected only in bulk-level data	Known candidate causal genes (bulk-level data)	<i>TSPAN14, GRN, CD2AP, APH1B, SLC39A13, FCER1G, CR1, NDUFAF6, TP53INP1</i>
	Known candidate causal genes (cell type-level data)	<i>FCER1G, GRN, CD2AP, APH1B, CR1</i>
Detected only in cell type-specific data	Known candidate causal genes (bulk-level data)	<i>RIN3, BIN1, CASS4, RABEP1, AL355353.1, KANSL1, SCIMP, USP6NL, PICALM, AL137789.1, FERMT2</i>
	Known candidate causal genes (cell type-level data)	<i>RIN3, PICALM, RABEP1, SCIMP, USP6NL, CASS4, BIN1, AL355353.1, JAZF1</i>
	Known bulk-Level Candidate Genes with New Cell Type Evidence	<i>AL137789.1, FERMT2</i>
	Novel cell type-level candidate causal genes (No or Limited Prior Evidence)	–
Detected in both bulk-level and cell type-specific data	Known candidate causal genes (bulk-level data)	<i>ACE, PRSS36, EGFR, SNX31, ARL17B, LRRC37A</i>
	Known candidate causal genes (cell type-level data)	<i>ACE, PRSS36</i>
	Known bulk-Level Candidate Genes with New Cell Type Evidence	<i>EGFR, SNX31, ARL17B, LRRC37A</i>
	Novel cell type-level candidate causal genes (No or Limited Prior Evidence)	<i>PABPC1</i>

novel candidate causal gene by SMR and colocalization analysis, with no or limited prior supporting evidence (Table 2).

To analyze interactions among the identified candidate causal genes, we constructed PPI networks as described in Methods. Our PPI analysis revealed interactions among the proteins encoded by 23 candidate causal genes, except for *PRSS36*, *CR1*, *TSPAN14*, *FCER1G*, *SLC39A13*, which showed no detectable interaction in the network, as illustrated in Fig. 3A. *EGFR*, identified in astrocytes, was the most highly connected node (n=7 interactions), suggesting it may serve as a central hub coordinating signals across multiple pathways (Fig. 3A). Other highly connected genes included *RIN3*, *CASS4*, and *PICALM*, all primarily detected in microglia, each with six interactions (Fig. 3A). Several strong interactions were observed among microglia-expressed genes, such as *BIN1* ~ *RIN3*,

CASS4 ~ *FERMT2*, and *CD2AP* ~ *PICALM*, indicating a dense microglial subnetwork (Fig. 3A).

To identify the enriched pathways and processes, we conducted enrichment analysis with multiple up-to-date databases, including KEGG 2021 Human, Reactome Pathways 2024, WikiPathways 2024 Human, GO Biological Process 2025, GO Cellular Component 2025, GO Molecular Function 2025, on the 26 candidate causal mRNA genes. Enrichment analysis showed that candidate causal genes are significantly associated with membrane organization, cell migration, and key signaling pathways including the ERK1/2 and PI3K/AKT cascades (GO Biological Process) (Fig. 3B and Table S6). Genes were also enriched in cellular components such as vesicles, early endosomes, focal adhesions, axons, and dendrites (Figure S7). Reactome analysis highlighted membrane trafficking, vesicle-mediated transport, and clathrin-mediated endocytosis as significantly enriched pathways (Fig. 3C).

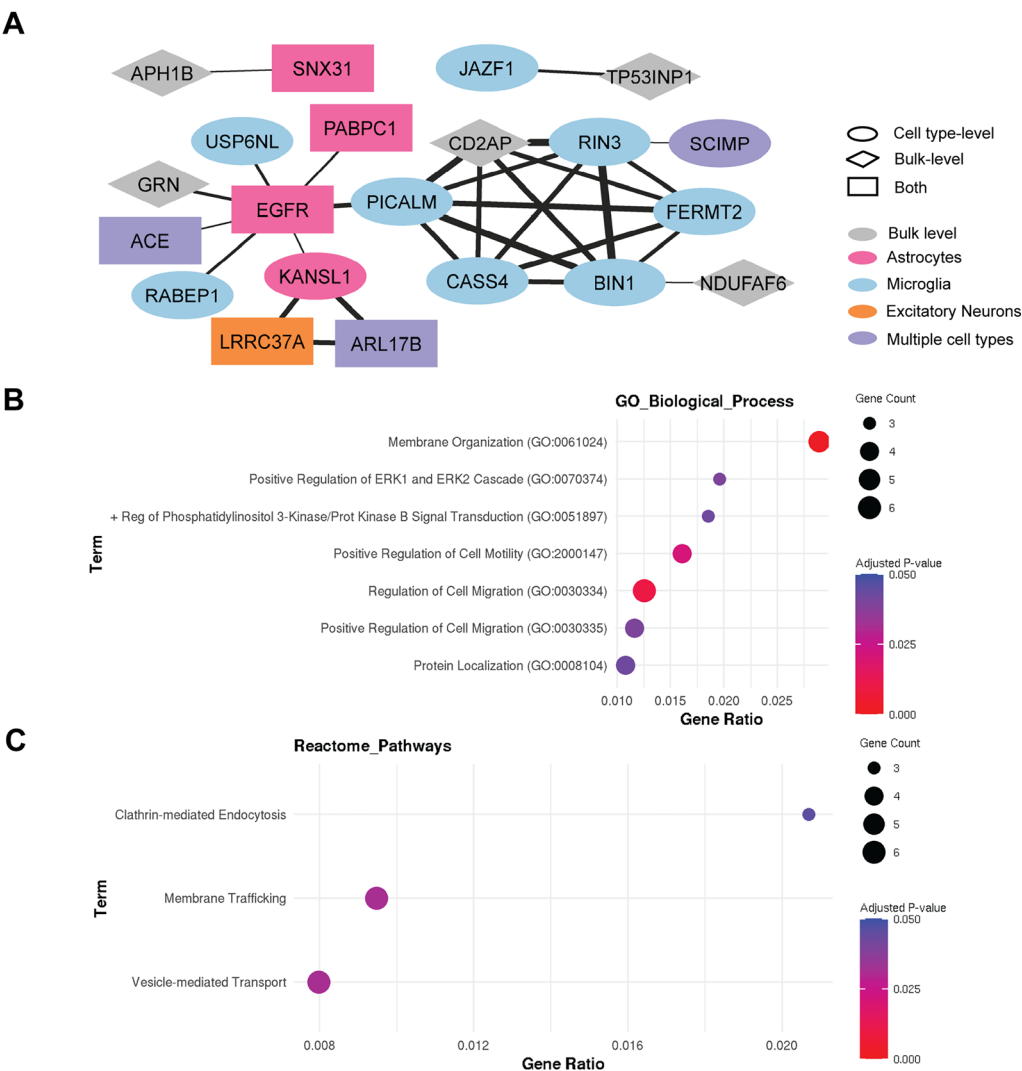


Fig. 3 Candidate causal genes network analysis and pathway enrichment. **A** STRING PPI network of candidate causal genes. Nodes represent proteins, while edges illustrate the interactions between them. The shape of each node was used to represent the detection context of candidate genes: ellipses indicate genes detected only in cell type-level datasets, diamonds represent genes found only in bulk-level datasets, and rectangles denote genes shared between both cell type-level and bulk-level analyses. The thickness of the edges corresponds to the combined interaction score from STRING. *ACE* and *SCIMP* were detected in both excitatory neurons and inhibitory neurons, while *ARL17B* was detected in both microglia and OPCs. **B** Pathway enrichment of candidate causal (mRNA) genes based on the Gene Ontology (GO) biological process category. **C** Pathway enrichment of candidate causal (mRNA) genes based on Reactome pathways

Visualization of colocalization for the novel astrocyte-specific candidate causal gene

We used eQTLPlot to visualize the colocalization between eQTL (Astrocyte specific eQTL from Mathys et al. [20]) and AD GWAS [4] signals for the novel candidate causal gene, *PABPC1*. As shown in Fig. 4A–C, *PABPC1* is indicated to be affected by the lead GWAS significant loci rs1693551 (GWAS P-value: 1.785e–08; Beta: 0.0459 from Bellenguez et al. [4] AD GWAS summary statistics data). Our analysis indicates that rs1693551 may also affect the other nearby gene *SNX31* (Fig. 2 and

Fig. 4B). We observed a tendency for eQTL to be over-represented in the lists of significant variants from the AD GWAS (p-value=1.12e–4 for *PABPC1* in astrocyte) (Fig. 4D). Congruous SNPs effect on the gene expression in astrocyte and AD risk were also observed for *PABPC1* (Fig. 4A, E, F). eQTLPlot P-value correlation analysis further confirms the colocalization between the *PABPC1* gene expression in astrocyte and AD risk as shown in Fig. 4E (r=0.85, p=1.19e–72). The variant rs1693551, with reference allele T and alternative allele C, was not identified as a new risk locus in the latest GWAS study

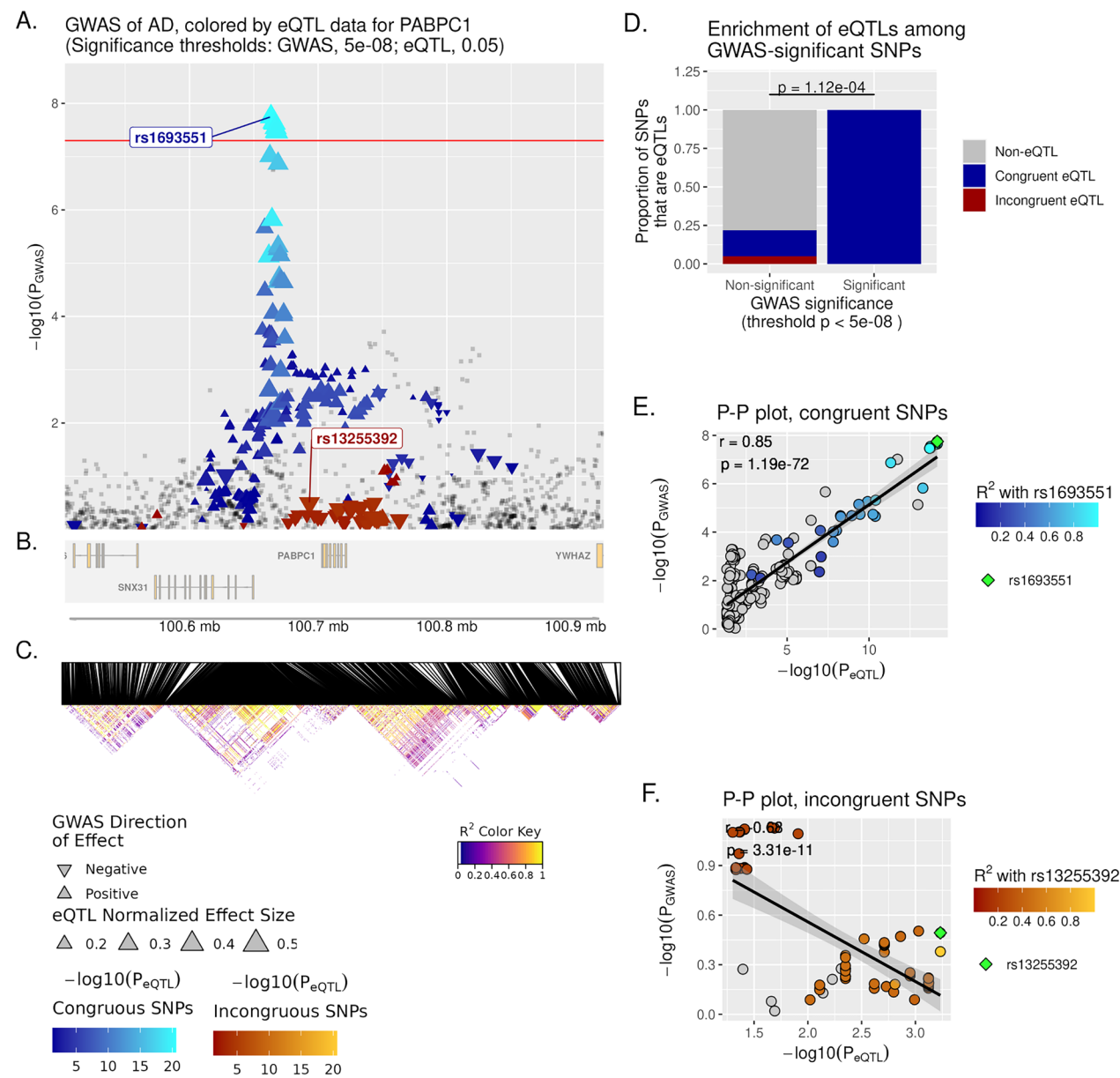


Fig. 4 eQTLPlot for colocalization between eQTLs for the gene *PABPC1* and a GWAS signal for AD. The GWAS dataset is from Bellenguez et al. [4] and the cell type eQTL dataset of astrocyte is from Mathys et al. [20]. **A** Shows the locus of interest, containing the *PABPC1* gene, with chromosomal space indicated along the horizontal axis. The position of each point on the vertical axis corresponds to the p-value of association for that variant with AD, while the color scale for each point corresponds to the magnitude of that variant's p-value for association with *PABPC1* expression. Variants with congruous effects are plotted using a blue color scale, while variants with incongruous effects are plotted using a red color scale. The directionality of each triangle corresponds to the GWAS direction of effect, while the size of each triangle corresponds to the effect size for the eQTL data. The default genome-wide p-value significance threshold for the GWAS analysis, $5e-8$, is depicted with a horizontal red line. **B** Displays the genomic positions of all genes within AD. **C** Depicts a heatmap of LD information of all *PABPC1* eQTL variants, displayed in the same chromosomal space as panels A and B for ease of reference ($R^2_{\text{min}}=0.1$, $LD_{\text{min}}=10$). **D** Depicts the enrichment of *PABPC1* eQTLs among GWAS-significant variants, while **E** and **F** depicts the correlation between P_{GWAS} and P_{eQTL} for *PABPC1* and AD, with the computed Pearson correlation coefficient (r) and p-value (p) displayed on the plot. For **E**, the analysis is confined only to variants with congruous directions of effect, while for **F** the analysis includes only variants with incongruous directions of effect. A lead variant is indicated in both **E** and **F**, and both are also labeled in **A**

[4]. However, our analysis reveals that it exceeds the genome-wide significance threshold, as illustrated by the Manhattan plot for chromosome 8 shown in Additional file 2: Figure S8. Additionally, we also observed colocalization of the shared causal variant for *PABPC1* gene expression and AD risk with eQTL datasets from Meta-brain (Additional file 2: Figure S9).

The MR and colocalization analyses identified a causal link between *PABPC1* gene expression in astrocytes and AD risk. To further explore this relationship, we examined *PABPC1* expression in both astrocytes and astrocyte subtypes, and its association with AD pathology, cognitive function and AD groups. Specifically, we utilized differential gene expression (DEG) results from a previous study [20] focused on the DLPFC region and applied multiple testing corrections. The findings, presented in Additional file 2: Figure S10, indicate that *PABPC1* expression in astrocytes is significantly associated with perceptual orientation, but not associated with AD diagnosis. Additionally, the expression of *PABPC1* in the astrocyte sub-type GRM3 shows a significant association with tangle density.

Enhancers harboring AD risk variants regulate cell-type-specific gene expression

Our results reveal that certain genes, such as *PABPC1*, were identified as candidate causal gene exclusively in one cell type, but not in other brain cell types. To explore the SNP and gene expression relationship, we examined genotype-dependent expression patterns and found a stepwise increase in *PABPC1* expression from genotypes TT to TC to CC, observed exclusively in astrocytes ($\text{FDR}=5.32 \times 10^{-12}$) (Figure S11). The nearby gene *SNX31* affected by the same SNP also exhibited a much stronger eQTL association in astrocytes ($\text{FDR}=8.97 \times 10^{-52}$) compared to excitatory neurons ($\text{FDR}=0.017$) and OPCs ($\text{FDR}=0.00038$) (Figure S12), suggesting a potential astrocyte-specific regulatory mechanism at this locus. This highlights that many candidate causal genes may be specific to a single cell type. To further understand this cell-type-specific effect, it is crucial to investigate how these variants influence gene expression and the underlying regulatory mechanisms. Enhancers are genomic regions that regulate gene expression, often in a cell-specific manner. A previous study [19] analyzed enhancer and promoter activity in human brain cell nuclei, revealing that genetic variants associated with brain traits and diseases exhibit cell-specific enhancer enrichment patterns. To determine if the cell-type-specific causal genes identified in our study are regulated by cell-type-specific enhancer activity, we analyzed a publicly available dataset, including ATAC-seq for open chromatin regions and ChIP-seq for active enhancers (H3K27ac) and promoters

(H3K4me3) for each brain cell type, as detailed in the Methods section.

As illustrated in Fig. 5, for the candidate causal gene *PABPC1* and *SNX31* in astrocytes, the associated disease variant is rs1693551 (chr8, hg19_position: 10,675,584 bp), which is located just 59 bp from the boundary of an astrocyte-specific enhancer (chr8, hg19_position: 101,675,643–101,676,301 bp) identified in the previous study [19]. Given its proximity to the enhancer boundary, it is possible that the enhancer region extends beyond what was detected, especially considering the dynamic nature of enhancers and technical limitations of current detection methods. Figure 5 shows that this enhancer, located downstream of the *PABPC1* gene and upstream of *SNX31*, is active only in astrocytes, evidenced by prominent H3K27ac and ATAC-seq peaks, while not active in other cell types. This suggests that the variant likely influences gene expression through a cell-type-specific enhancer, which may explain why *PABPC1* was detected as a causal gene exclusively in astrocytes and why *SNX31* expression was more affected by the genotype of this locus in astrocyte than other cell types.

Furthermore, a previous Hi-C (High-throughput Chromosome Conformation Capture) dataset from 11 postmortem female brain samples (prefrontal cortex), including 4 Alzheimer's disease (AD) cases, 4 aged cognitively normal controls, and 3 young cognitively normal controls, revealed that *SNX31*, rs1693551, and *PABPC1* are located within a chromatin loop present in both young and aged control brains but notably absent in AD brains [35] (Figure S13). Increasing dosage of the rs1693551 risk allele correlates with elevated expression of both *SNX31* and *PABPC1*, suggesting that this SNP may influence transcription within a shared regulatory domain.

Druggability analysis and drug/compound prediction

To identify druggable genes from our candidate causal genes, we categorized them based on a prior drug tier classification [27]. Tier 1 includes targets of approved drugs and clinical candidates; Tier 2 includes targets with known drug-like interactions or high similarity to approved drug targets; and Tier 3 includes proteins with distant similarities to drug targets or those in key druggable families, as mentioned in the Methods. As detailed in Additional file 1: Table S7, we identified three candidate causal genes, *EGFR*, *ACE*, and *APH1B*, as Tier 1 druggable, and three genes, *GRN*, *PRSS36*, and *CR1*, as Tier 3 druggable. The remaining candidate causal genes were not classified as druggable based on the previous study [27]. For these non-druggable genes, we used EpiGraphDB to prioritize potential alternative drug targets within the same PPI network. We identified directly

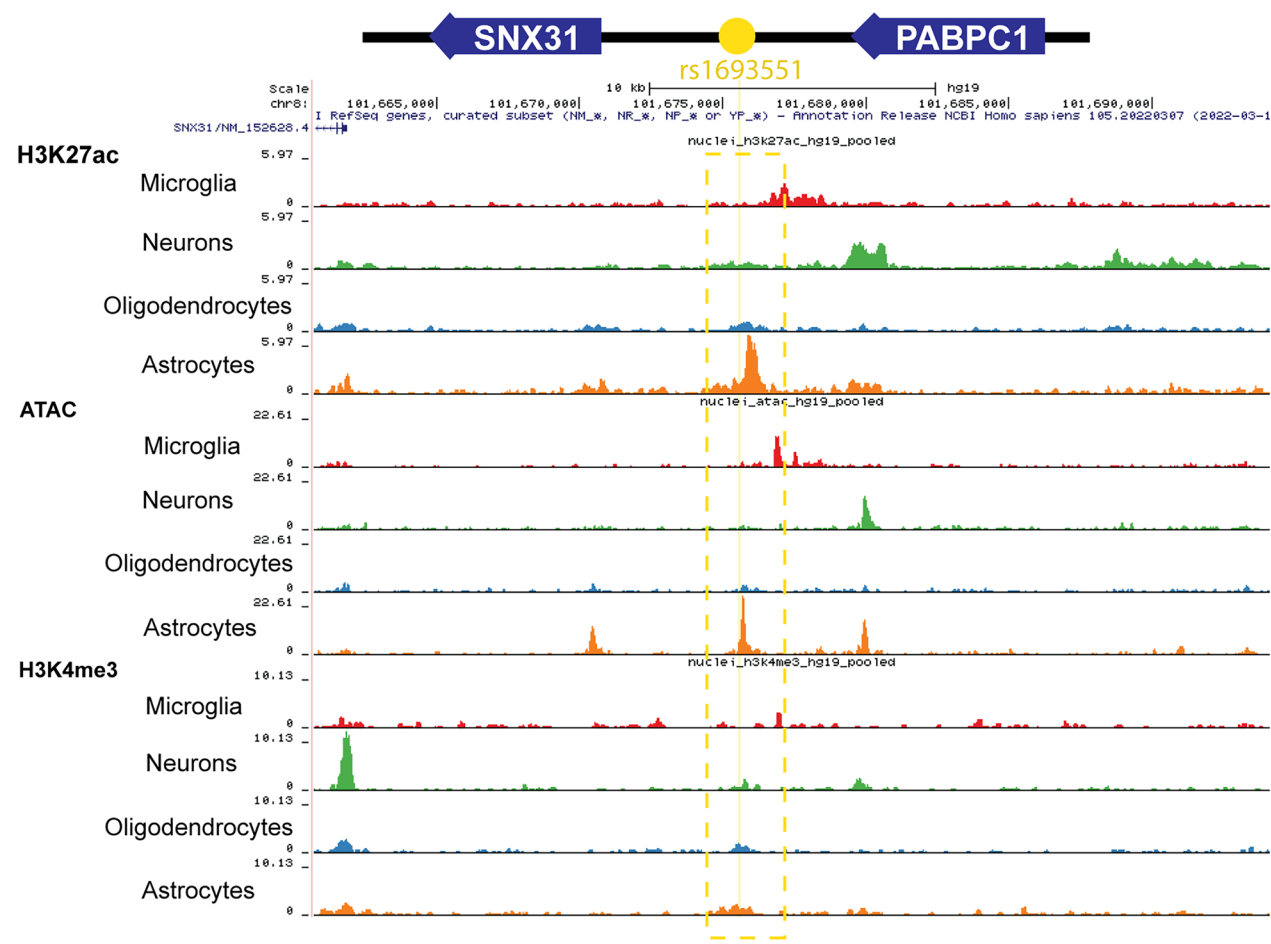


Fig. 5 Brain cell-type-specific chromatin profiles by UCSC Genome Browser (hg19). A H3K27ac and ATAC-seq data for *PABPC1*, showing active enhancer regions and open chromatin specific to astrocytes, with a yellow vertical line marking the location of the associated disease variant and a dashed square showing the region of active enhancer

AD related interacting genes with Tier 1 druggability using PPI networks from IntAct and STRING databases, shown in Additional file 1: Table S7.

To identify drugs targeting the causal genes identified in this study and to broaden the scope of potential drug targets, we conducted a drug/compound enrichment analysis using DSigDB. This analysis aimed to find potential drugs for all the target genes, which include both the druggable causal genes identified in this study and directly interacting genes with Tier 1 druggability, as detailed in Additional file 1: Table S7. The results of the enrichment analysis are presented in Additional file 1: Table S8. We focused on drugs with an adjusted p-value of less than 0.01 and selected the top 10 most significant potential drugs/compound based on their adjusted p-value (Additional file 1: Table S8 and Fig. 6A). Figure 6A presents the drugs grouped by gene ratio (the percentage of target genes overlapping with the drug gene set). Within each group, the drugs are ranked by

their adjusted p-value significance. The results highlight that 3-(1-methylpyrrolidin-2-yl)pyridine targets the highest number of genes, with 16 target genes including *EGFR*, *ACE*, *MAPK1*, *TNFRSF1A*, *EEF2*, *ADRB2*, *CD4*, *APP*, *TFRC*, *ITGAL*, *PLD1*, *FYN*, *PIK3CA*, *RAF1*, *TP53* and *VEGFA* (Additional file 1: Table S8). In the second group, Dinoprostone is the most significant drug, targeting 14 genes. In the third group, Imatinib mesylate is the most significant drug, targeting 13 genes, followed by histamine. Imatinib mesylate is detected as the most significant drug across groups. These top 10 enriched drugs (Fig. 6A) show promise for therapeutic applications in AD and need further investigation.

To illustrate the interactions between drugs and target genes (both causal genes identified in this study and directly interacting genes (AD related) with Tier 1 druggability), we constructed an interaction network using Cytoscape, as shown in Fig. 6B. This network highlights that Tier 1 druggable genes, such as *EGFR* (targeted by all

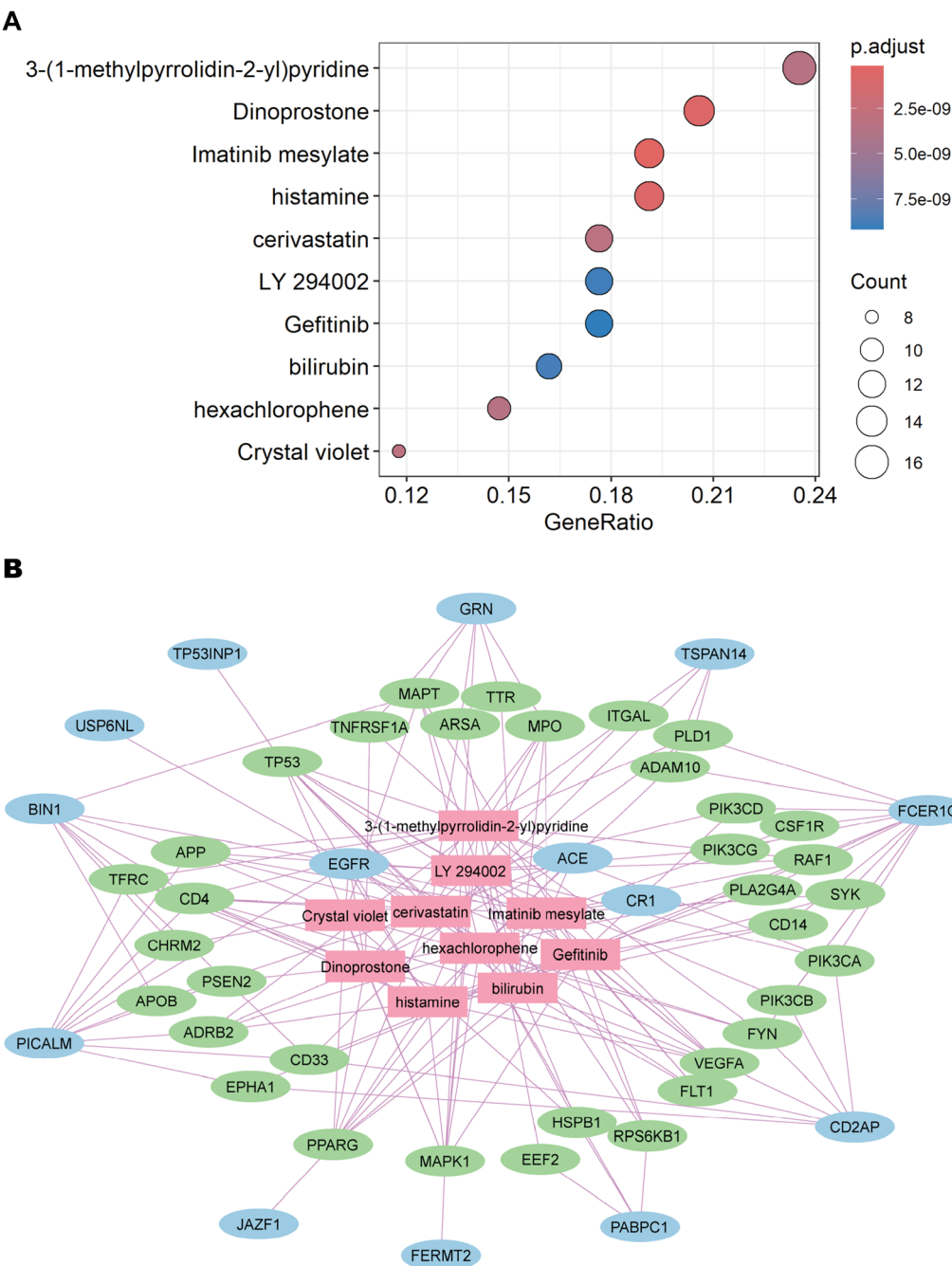


Fig. 6 Potential drugs enrichment analysis and gene-drug interaction network. **A** Top 10 enriched drug/compounds based on DSigDB predictions. **B** Interaction network illustrating connections between enriched drugs/compounds and target genes. Blue circles indicate druggable/non-druggable causal genes identified in this study, green circles represent druggable interacting genes linked to non-druggable causal genes, and pink nodes denote the top 10 enriched drugs/compounds

top 10 drugs) and *ACE* (targeted by 5 of the top 10 drugs) (Additional file 1: Table S8 and Fig. 6B), are directly targeted by multiple drugs. Additionally, the Tier 3 drug-gene *CR1* is directly targeted by Imatinib mesylate. In the network, druggable and non-druggable causal

genes are represented by blue circles; interacting genes are shown in green circles, and drugs/compounds are depicted in pink (Fig. 6B). The central area of the network features drugs and Tier 1 druggable genes, indicating direct targeting, while the surrounding groups represent

interacting genes and non-druggable causal genes, which are indirectly targeted through these interactions. This visualization demonstrates the role and significance of the top 10 drugs in targeting multiple causal genes, both directly and indirectly (Fig. 6B).

Discussion

Many disease-associated loci exert effects that are specific to cell types [11, 14, 36, 37]. Brain diseases are influenced by genetic effects that are specific to both cell types and brain regions [11, 14, 38]. Previous GWAS studies often identify risk variants that impact disease phenotypes by regulating genes in specific tissues, yet the precise cell types involved are often not well characterized [10, 39]. Our study addresses this knowledge gap by using brain single-cell eQTL data to reveal how genetic variants impact AD at the cellular level, offering crucial insights into cell-type-specific regulation driving the disease. In this study, we combined data from five recent AD GWAS with two cell type-level eQTL datasets from single-cell RNA sequencing and one bulk tissue eQTL dataset from a prior meta-analysis. Through SMR and colocalization analyses, we identified candidate causal genes at both bulk and cell-type levels, uncovering novel genes and confirming known ones. We investigated gene regulation in specific cell types by analyzing enhancer activity using previous H3K27ac, ATAC-seq and Hi-C data. Network and pathway enrichment analyses provided additional insights into the biological relevance of these genes. To facilitate drug repurposing for AD, we performed a drug/compound enrichment analysis using the DSigDB, mapping drug interactions with both causal and interacting druggable genes. This integrated approach highlights the importance of cell-type specific functional evidence in genetic research, revealing how AD GWAS variants contribute to disease through cell-specific gene expression. By examining genetic effects at the cellular level, we gain clearer insights into AD molecular mechanism and identify promising targets for drug discovery.

In recent years, there has been growing recognition of the context-specific nature of eQTLs, extending from tissue types to functional, environmental, and cellular contexts [11, 14, 40–42]. Our study underscores the critical value of cell-type-specific eQTL datasets in identifying candidate causal genes for AD. Specifically, we identified 12 genes exclusively as candidate causal genes within the cell-type eQTL datasets (Figure S6). This finding highlights the limitations of bulk tissue analyses, which often aggregate signals across various cell types and may miss gene-regulatory effects that are specific to cellular contexts. By focusing on cell-type-specific eQTL data, we can uncover gene associations that are masked when only bulk tissue data is used. Furthermore, of the 19 candidate

causal genes identified through cell-type-specific eQTL datasets, 16 were found to be causal in only one cell type (Fig. 2 and Table 1). This cell-type specificity highlights the importance of considering cellular heterogeneity in genetic studies of complex diseases like AD.

In our analysis, *PABPC1* emerged as a novel candidate causal gene for AD, highlighting its potential role in disease mechanisms. Specifically, the MR and colocalization analyses identified a causal link between *PABPC1* gene expression in astrocytes and AD risk. We found that *PABPC1* expression in astrocytes is significantly linked to perceptual orientation and significantly associated with tangle density in the GRM3 astrocyte subtype. Although we did not observe a significant association between *PABPC1* expression and AD diagnosis in astrocytes, previous studies have reported its upregulation in other brain regions, such as the parahippocampal gyrus region [43] (Figure S14). *PABPC1* is known to bind tau proteins [44]. It also regulates translation and mRNA stability [45]. Additionally, *PABPC1* is involved in stress granules and RNA splicing, critical for managing cellular stress and maintaining protein synthesis [46]. Its associations with neurofilament light chain (NF-L) [47], along with its co-localization with small tau inclusions in tauopathy [48], underscore its relevance in AD pathology. These findings warrant further investigation into *PABPC1* as a potential therapeutic target. The AD risk loci rs1693551, which achieved GWAS significance only in the latest AD GWAS summary statistics [4], has been less studied. It is the leading GWAS locus associated with the expression of the causal genes *SNX31* and *PABPC1* in astrocytes, underscoring its potential significance in AD. This highlights the need for further investigation into its role and relevance in the disease.

In our results, 16 of the 19-cell type level candidate causal genes were found to be causal in only one specific cell type (Fig. 2 and Table 1). Previous research indicates that cell-type-specific enhancers harboring AD-risk variants can drive such cell type-specific gene regulation [19]. For example, while *PICALM* and *BINI* are expressed in multiple cell types, they contain microglia-specific enhancers with AD-risk variants [19]. Although direct enhancer–promoter interactions remain unconfirmed due to the absence of PLAC-seq data in astrocytes [19], bulk-level Hi-C data from control brains reveal a chromatin loop encompassing rs1693551, *SNX31*, and *PABPC1*. Notably, this loop is absent in AD brains, suggesting that its loss may disrupt regulatory interactions and contribute to the dysregulation of these genes. eQTL analysis indicates a regulatory effect of the SNP on both genes in astrocytes. This indicates that regulatory effect could be from dynamic chromatin configurations or involvement in shared transcriptional hubs. These findings support

a potential pleiotropic role for rs1693551, mediated by higher-order chromatin architecture, and underscore the importance of 3D genome organization in mediating gene regulatory changes in AD. In addition to microglia, which are well-known for their roles in AD, our study highlights the importance of astrocytes. We provide more molecular evidence showing that astrocytes are critically involved in AD through specific gene expression and enhancer activity associated with AD-risk variants.

Our analysis identified 19 candidate causal genes at the cell type level, with 16 showing cell type-specific effects. While this may reflect true biological specificity, However, it's essential to consider technical factors such as single-cell RNA-seq data quality, gene expression variability, sample size, and the proportion of each cell type within the data. Single-cell RNA-seq, while powerful, is prone to technical noise, including amplification biases, dropout events, and variability between individual cells, making it less reliable compared to bulk RNA-seq data. Previous studies have found that genes with cell type-specific eQTLs often have higher expression variability, making them easier to detect in eQTL analyses [49–51]. Additionally, the ability to detect these effects depends on factors like the number of samples and how well each cell type is represented in the data. With fewer cells in certain cell types, such as vascular cells, it becomes harder to identify causal genes. Thus, while our findings suggest some biological specificity, the technical factors like expression variability and differences in cell type representation likely play a significant role in detecting these associations. Adequate sample size is critical for reliable eQTL detection, particularly when evaluating allele-specific effects. In this analysis, although *FERMT2* was identified as a potential causal gene in immune cells, the number of individuals carrying the risk allele was much lower than those without it (Figure S15). This imbalance may limit statistical power for eQTL analysis and impact the detection of causal gene. Future studies with larger sample sizes, better brain sample quality, and improved single-cell sequencing techniques will be essential for separating these biological and technical influences and better understanding the role of these genes in different cell types.

Our DSigDB enrichment analysis identified several drugs/compounds with potential therapeutic relevance for AD, such as Imatinib mesylate, histamine, Dinoprostone, 3-(1-methylpyrrolidin-2-yl)pyridine, Gefitinib, Crystal violet, cerivastatin, and hexachlorophene. Imatinib mesylate was highlighted as the most significant drug (Additional file 1: Table S8). Imatinib mesylate is notable for its role as a tyrosine kinase inhibitor and has been shown to reduce A β production in various experimental models [52]. Research suggests it may be

effective in treating neurodegenerative disorders, including AD [53]. However, further studies are needed to fully understand its effects on the brain, particularly its ability to cross the blood–brain barrier. Some research has explored how imatinib interacts with brain transporters such as breast cancer resistance protein and P-glycoprotein [54], which is important for optimizing its use in neurodegenerative diseases. 3-(1-methylpyrrolidin-2-yl)pyridine (Nicotine) stands out for targeting the highest number of analyzed genes. Nicotine, an alkaloid in tobacco, functions by activating nicotinic acetylcholine receptors (nAChRs), which are widely expressed throughout the nervous system [55]. It has dual effects on oxidative stress and neuroprotection [56], suppresses neuroinflammation [57], and prevents A β aggregation [58]. Despite these benefits, its use in AD is limited by cardiovascular risks [59], addiction and negative associations with smoking [60]. However, Nicotine's gene targeting profile found in this study suggests it could impact multiple pathways involved in AD, potentially offering a therapeutic approach through nicotinic derivatives that mitigate these adverse effects.

There are several limitations in this study. The study analyzed data from various brain regions across multiple datasets, including the cortex from the bulk metabrain eQTL dataset, the DLPFC region from the Mathys_2023 snRNA eQTL dataset, and a range of regions such as the temporal cortex, white matter, and PFC from the Bryois 2021 snRNA eQTL dataset. The variability in brain regions might limit the generalizability of our findings, as genetic effects can be region-specific. A limitation of this study is the partial sample overlap between some of the snRNA-seq eQTL datasets and the GWAS datasets used in colocalization and SMR analyses. Specifically, the cell type-level eQTL datasets are partly derived from the ROSMAP cohort, which also contributes to four of the five GWAS datasets included in this study (Table S1). However, it is important to note that the ROSMAP-based eQTL datasets comprise only a few hundred individuals, whereas the GWAS cohorts typically involve tens to hundreds of thousands of participants. As such, the degree of overlap represents a small fraction of the total sample size and is unlikely to meaningfully bias the integrative analyses. Moreover, most gene prioritization results were observed in the Bellenguez et al. [4] GWAS, which does not include ROSMAP samples in its stage 1 discovery phase used in this study, supporting the robustness of our findings. Nonetheless, we acknowledge the possibility of minor inflation in test statistics due to limited overlap and have noted this accordingly. However, the Meta-Brain bulk eQTL dataset, which includes 2,683 samples, has a greater overlap with the GWAS datasets (Tables S1 and S2), which may lead to more inflation in the test

statistics. Therefore, further validation of these candidate causal genes is required. Additionally, some GWAS datasets used in this study incorporate proxy cases, individuals with a family history of AD but no clinical diagnosis, which may introduce phenotype heterogeneity. While this approach increases power for genetic discovery, it could also dilute associations with clinical AD. Future studies incorporating sensitivity analyses limited to clinically diagnosed AD cases will be important for refining causal inference.

Also, the GWAS and eQTL datasets primarily included individuals of European ancestry, which limits the generalizability of the findings to other ethnic groups. Additionally, our analysis was limited to cis-eQTLs, which reflect direct effects on genes. Cis-eQTLs do not capture the full spectrum of genetic influences, as trans-eQTLs could reveal downstream gene sets and pathways affected by disease variants. Future studies should explore available cell-specific trans-eQTL data to better understand the causal effects of genetic variants acting in trans. Furthermore, future research should use independent snRNA eQTL datasets for validation. While our study identified potential drug targets through enrichment analysis, their clinical efficacy remains unconfirmed. Experimental validation and clinical trials are necessary to establish their therapeutic potential. Moreover, since the candidate causal genes were identified from brain tissue data and in drugs that face challenges in crossing the blood–brain barrier, further investigation is needed to evaluate the viability of these targets for drug development. Lastly, the identification of *PABPC1* as a novel astrocyte-specific candidate gene is compelling, but functional validation is essential to confirm its regulatory role in AD. Quantitative PCR and immunohistochemistry could be used to validate the expression of *PABPC1* specifically in astrocytes from AD and control brain tissues. For regulatory effect validation, CRISPR interference (CRISPRi) targeting the rs1693551 enhancer region could be applied in iPSC-derived astrocytes to assess its causal effect on *PABPC1* and *SNX31* expression. In addition, assays such as chromatin conformation capture in astrocyte-specific models would help determine whether enhancer–promoter loops involving rs1693551 are disrupted in AD.

Our analysis identified both novel and established candidate causal genes, elucidating their roles in AD molecular mechanisms and highlighting the significance of cell-type specificity in gene expression regulation and enhancer activity.

Abbreviations

AD	Alzheimer's Disease
eQTLs	Expression Quantitative Trait Loci
GWAS	Genome-Wide Association Study
SMR	Summary Data-Based Mendelian Randomization
COLOC	Bayesian Colocalization

LOAD	Late-Onset Alzheimer's Disease
UKBEC	The UK Brain Expression Consortium
GTEx	Genotype-Tissue Expression Consortium
DLPFC	Dorsolateral Prefrontal Cortex
PFC	Prefrontal cortex
OPCs	Oligodendrocyte Progenitor Cells
TMM	Trimmed mean of M-values
CPM	Counts per million
PCs	Principal components
HEIDI	Heterogeneity in Dependent Instruments
PPs	Posterior probabilities
PPI	Protein–protein interaction
LD	Linkage disequilibrium
DSigDB	The Drug Signatures Database
DEG	Differential Gene Expression
Hi-C	High-throughput Chromosome Conformation Capture

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12967-025-06739-1>.

Additional file 1: Table S1. Alzheimer's disease GWAS studies. Table S2. Brain cortex region cis-eQTL datasets. Table S3. SMR and Coloc analysis results for metabrain eQTL and AD GWAS summary statistics. Table S4. SMR and Coloc results for Bryois cell type specific eQTL and AD GWAS summary statistics. Table S5. SMR and Coloc results for Mathys cell type specific eQTL and AD GWAS summary statistics. Table S6. Pathway enrichment of candidate causal genes. Table S7. Druggability of candidate causal genes. Table S8. Drug/compound enrichment analysis results.

Additional file 2: Figure S1. SMR beta value and significance for candidate causal genes from SMR and colocalization analysis. Figure S2. SMR beta value and significance for candidate causal genes from SMR and colocalization analysis. Figure S3. SMR beta value and significance for candidate causal genes from SMR and colocalization analysis. Figure S4. SMR beta value and significance for candidate causal genes from SMR and colocalization analysis. Figure S5. SMR beta value and significance for candidate causal genes from SMR and colocalization analysis. Figure S6. Venn diagram comparing genes between two snRNA (Bryois and Mathys) and bulk metabrain eQTL datasets. Figure S7. Pathway enrichment of candidate causal genes based on the Gene Ontology (GO) cellular component category. Figure S8. Manhattan plot of AD GWAS [4] on chromosome 8. Figure S9. eQTLPlot for colocalization between eQTLs for the gene *PABPC1* and a GWAS signal for AD. Figure S10. DEGs detection of *PABPC1* with pathology, cognitive function and AD. Figure S11. Boxplot of *PABPC1* gene expression across rs1693551 genotypes in different cell types. Figure S12. Boxplot of *SNX31* gene expression across rs1693551 genotypes in different cell types. Figure S13. Chromatin Loop Analysis and Interaction Visualization. Figure S14. *PABPC1* gene expression fold changes plot from Agora website. Figure S15. Boxplot of *FERMT2* gene expression across rs17125924 genotypes in immune cells.

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Author contributions

S. L. performed the data analysis and wrote the manuscript. All other authors contributed to manuscript revisions and reviewed the manuscript.

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Availability of data and materials

GWAS summary statistics for AD were downloaded from https://ftp.ebi.ac.uk/pub/databases/gwas/summary_statistics/GCST007001-GCST008000/GCST007511/, https://ftp.ebi.ac.uk/pub/databases/gwas/summary_statistics/GCST013001-GCST014000/GCST013197/, http://ftp.ebi.ac.uk/pub/databases/gwas/summary_statistics/GCST90027001-GCST90028000/GCST90027158/, http://ftp.ebi.ac.uk/pub/databases/gwas/summary_statistics/GCST90012001-GCST90013000/GCST90012877/, http://ftp.ebi.ac.uk/pub/databases/gwas/summary_statistics/GCST007001-GCST008000/GCST007320/.

Publicly available summary statistics of metabrain eQTLs was obtained from MetaBrain website (<https://www.metabrain.nl/>). Bryois Single cell eQTL dataset was obtained from <https://doi.org/10.5281/zenodo.5543734>. Mathys et al. [20] snRNA dataset from ROSMAP cohort (downloaded from Synapse: syn52293442). The publicly available dataset, including ATAC-seq, which identifies open chromatin regions, and ChIP-seq, which marks active enhancers (H3K27ac) and promoters (H3K4me3) for each brain cell type, accessed through the UCSC genome browser session (hg19) at: https://genome.ucsc.edu/s/nottalexi/glassLab_BrainCellTypes_hg19. The brain Hi-C data is accessed through <http://menglab.pub/hic/>.

Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Declarations

Ethics approval and consent to participate

This study used publicly available datasets, no ethics approval and consent to participate was not required.

Consent for publication

This study used publicly available datasets. All data were anonymized, and no information that could reveal the identity of participants was used. Therefore, consent for publication from individual participants was not required.

Competing interests

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Author details

¹Center for Neuroimaging, Department of Radiology and Imaging Sciences, Indiana University School of Medicine, 550 N. University Blvd., Indianapolis, IN 46202, USA. ²Indiana Alzheimer's Disease Research Center, Indiana University School of Medicine, 355 W. 16th Street, Goodman Hall, Suite 4100, Indianapolis, IN 46202, USA. ³Sungkyunkwan University, Seoul, Republic of Korea. ⁴Department of Biomedical Informatics, College of Medicine, The Ohio State University, Columbus, OH 43210, USA. ⁵Department of Neurological Science, Rush Alzheimer's Disease Center, Rush University Medical Center, Chicago, IL 60612, USA. ⁶Department of Neuroscience, Mayo Clinic, Jacksonville, FL 32224, USA. ⁷Department of Neurology, Mayo Clinic, Jacksonville, FL 32224, USA. ⁸Center for Computational Biology and Bioinformatics, Indiana University School of Medicine, 340 West 10th Street, Fairbanks Hall, Suite 6200, Indianapolis, IN 46202, USA.

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